AE460B Spacecraft Design

TM05: Operations Subsystem Design

Destiny

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This technical memorandum presents the preliminary design of the Operations Subsystem for our Solar Gravitational Lens (SGL) spacecraft mission. The operations architecture is developed based on the unique challenges of operating a spacecraft at distances greater than 650 AU, where communication delays can reach up to several days. Since we have these delays, the spacecraft cannot rely on real-time ground control and must incorporate a high level of onboard autonomy. This memo outlines the mission-driven operational requirements, telemetry and telecommand architecture, subsystem interfaces, operational phases, staffing concepts, and risk considerations that shape our overall operations design.

The proposed concept includes a Mission Operations Center (MOC), Deep Space Network ground support, and structured telemetry processing and command generation procedures. This memo describes how telemetry flows from the spacecraft to the ground and how commands are generated, validated, and uplinked. It also defines operational phases such as LEOP, commissioning, and routine science operations. Emphasis is placed on automation, time-tagged commanding, anomaly response procedures, and risk mitigation strategies to ensure reliable long-duration operations. Overall, the selected architecture balances onboard autonomy with ground oversight to support sustained science operations at extreme deep-space distances.

Nomenclature

ADCS	=	Attitude Determination and Control System
AU	=	Astronomical Unit
BIT	=	Built-In Testing
CCSDS	=	Consultative Committee for Space Data Systems
COM	=	Communications subsystem
DSN	=	Deep Space Network
E2E	=	End-to-End
EPS	=	Electrical Power Subsystem
FDIR	=	Fault Detection, Isolation, and Recovery
GEO	=	Geostationary Earth Orbit
IR	=	Infrared
IR TCC	=	Infrared Telecommand and Control
JPL	=	Jet Propulsion Laboratory
kB/s	=	Kilobytes per second (written as KB/s in text)
LEO	=	Low Earth Orbit
LEOP	=	Launch and Early Operations Phase
MOC	=	Mission Operations Center
NASA	=	National Aeronautics and Space Administration
OBC	=	Onboard Computer
OBDH	=	Onboard Data Handling
PCb	=	Proxima Centauri b
RF	=	Radio Frequency
RTG	=	Radioisotope Thermoelectric Generator
s/c	=	Spacecraft
SGL	=	Solar Gravitational Lens
TMI	=	Trans Mars Injection
TM	=	Telemetry (also written as TLM)
TWTA	=	Traveling-Wave Tube Amplifier
UTC	=	Coordinated Universal Time
WASC	=	West Australia Space Center
X-band	=	X-band (radio-frequency band)
XIPS	=	Xenon Ion Propulsion System
Δv	=	Change in velocity (delta-v)
v_∞	=	Hyperbolic excess velocity
C_3	=	Characteristic energy ($C_3 = v_\infty^2$)
e	=	Orbital eccentricity
R_J	=	Jupiter radius
μrad	=	Microradian (pointing accuracy unit used in text)

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I. Mission Context and Operations Requirements

A. Mission scenario

The Solar Gravitational Lens (SGL) mission is to be operated via a single spacecraft performing a distance flyby to observe the Einstein ring forming around our sun, and to observe the nearby planet Proxima Centauri b (PCb) with high resolution. The range in which we will be able to observe PCb is at a distance of 650-900 AU from our sun. Before 650 AU, the gravitational pull of the sun would not bend light far enough that the ring would be visible. After passing 900 AU, the ring will be too obscured by the sun's halo, and imagery would not be effective [1]. To reach this distance within a feasible time frame, we must acquire a very large v_{inf} exiting the solar system. To do so, we perform a Trans Mars Injection (TMI) to build speed reaching Jupiter, where we pick up more speed and redirect towards the sun. Once there, we perform an Oberth Maneuver to gather the final gravity-assisted acceleration on our way out of the solar system. Once at perihelion, a solar sail is activated to garner even more speed. The timeline for each section and its effects is as follows:

Earth Departure: From GEO, the TMI maneuver provides $\Delta v = 6.73$ km/s to achieve $v_\infty = 8.79$ km/s ($C_3 = 77.28$ km^2/s^2) for the Jupiter transfer.

Jupiter Flyby: The spacecraft arrives at Jupiter with $v_\infty = 5.64$ km/s and performs a powered flyby at 1.0 R_J altitude. The Oberth maneuver ($\Delta v = 1.12$ km/s) places the spacecraft on a heliocentric orbit with aphelion at Jupiter (5.2 AU) and perihelion at 0.046 AU ($e = 0.9825$).

Solar Oberth: At perihelion, the spacecraft velocity reaches 195.53 km/s (local escape velocity: 196.39 km/s). The solar Oberth maneuver ($\Delta v = 36.34$ km/s) achieves escape at $v_\infty = 123.25$ km/s (26.0 AU/year), reaching 650 AU in 24.98 years. The Oberth effect saves 86.9 km/s compared to a deep-space burn.

Science Phase: Once having reached 650 AU, only axial maneuvers will be performed to handle stellar wobble, and the velocity of the craft is not expected to diminish by any non-negligible margin. It will take 9.60 years for the spacecraft to reach 900 AU, where the payload will enter the decommissioning phase.

Maneuver	**ΔV (km/s)**	**% of Total V**	**Cumulative Time**
Trans-Jupiter Injection (GEO)	6.73	5.46	0 yr
Jupiter Oberth	1.12	0.908	2.73 yr
Solar Oberth (0.046 AU)	36.34	29.5	4.85 yr
Science Phase (650 AU)	86.91	100	29.83 yr
Decommissioning Phase (900 AU)	0	100	39.43 yr
Total	**123.25**	100	**39.43 yr** to 900 AU

Table 2 Mission Δv budget and timeline.

The ground segments for the mission require constant communication with the spacecraft, and by using IR TCC, we must choose locations that are without much weather to suppress atmospheric interference. 4 ground sites have been chosen for this mission. In previous deep space missions (Voyager), 3 ground sites were apt for the mission, but since downlink is extremely important for the purpose of the flight, a redundant ground site was added.

Location	**Longitude (Degrees East)**	**$\Delta Degree$**	**Standard Weather Conditions**
Ortucchio, Italy	13.604	0	Occasional light clouds
Mingenew-Nangetty, West Australia	115.35	101.74	Dry clear skys
Haleakala, Hawaii	203.75	88.4	Cloudless above peak
Cerro Chajnantor, Chili	292.26	88.51	Occasional rain below peak

Table 3 Ground Station locations, longitudinal placements, and standard weather expectations

Each of these ground station sites is already home to the housing of RF and X-band telecommunication stations, making the introduction of our ground site hardware an easy adaptation, given the time to implement. These locations were also chosen for their light weather conditions. In the case of The Haleakala Observatory, and Cerro Chanjnantor's Atacama Observatory, their placements at the peaks of a volcano and mountain allow them to avoid nearly all rain fade and other similar intrusions [2, 3]. In the case of the West Australia Space Center (WASC), the environment of West Australia being arid means there is rarely any weather to contend with [4]. Finally, the Fucino Space Center tends to be the most volatile when it comes to weather; its change in altitude compared to its surroundings leaves it slightly less vulnerable given its location, making it the better of competing choices in the area [3]. These four were chosen based on their weather convenience, but also because of their placement around the globe. If rain fade were not an issue, three ground stations would have served to provide 24/7 communication with the spacecraft. The fourth round station was introduced as a redundancy to ensure we have continuous support for the spacecraft for the decades it will be traveling.

Because of the great distance to the spacecraft, by the time the s/c reaches the science phase (650 AU), it will take 3.75 days for us to receive data. Once it reaches the decommission phase, that becomes 5.20 days. This extraordinary distance means that it is not feasible for us to perform all operations manually. As a result, a large portion of spacecraft commanding will need to be done by autonomous processing. Many available technologies implement machine learning into spacecraft software [5]. Besides instances like fault repair, repatriating, and other critical operations that are judged to be time-dependent, the majority of non-critical commanding can still be done by ground-driven controllers. An example of this is maneuver uploads. Because there are no exotic gravity perturbations to consider, like in orbiting spacecraft, predicting the trajectory of the s/c days ahead is fairly simple. As a result, we will be able to receive position data from the s/c and plot its course, then plan future maneuvers to take place by the time they have been received by the s/c with little risk of failure. This allows us to avoid the risk of mistakes from autonomous maneuver planning, which has not been made a reliable technology at such a scale.

B. Operations Subsystem Definition and Scope

The operations subsystem is the most critical subsystem to the operations of the spacecraft. This is the system that connects the ground and flight operations together, and is used to plan, command, and monitor the mission. The ops subsystem is utilized in mission planning and scheduling through the creation of timelines (short to long term), holding to constraints like power and thermal, and scheduling spacecraft modes depending on the phase. These spacecraft modes take place at different stages; some like the downlink is constantly running, whereas the science mode is only active for a section of the overall mission. In order to monitor the spacecraft, the ops subsystem keeps track of telemetry from power, thermal, ADCS, propulsion, OBC, and other systems, and compares their current values to limits which will notify controllers if those limits are exceeded. The control side of this also sees commands being sent and activated. Within the context of monitoring, that control can take the form of payload or bus safeing, or changing between set modes (charging or discharging batteries). The generation of commands is done in advance of events such as maneuvers, and are set in the schedule to be automatically sent via time-tagged commanding. Validation that these procedures were completed is done through simulation and telemetry analysis.

When coordinating with ground systems, the ops subsystem must coordinate between the two sides and identify what had failed to pass between them due to weather or visibilty constraints. This is also where the handoff of status between ground sites is done when the spacecraft goes out of range of one. Uplink and downlink budgets must be tracked here so we can prioritize bus over payload data. When an issue does occur, anomaly detection and response is needed, including FDIR trips and ground alarms. The ops subsystem is what decides when to switch between safe, recover, or reconfiguration modes based on the situation. In the long term, the ops subsystem updates limits based on long term trends, and handles the overall changes as a result of single failures (such as TWTA failures, or XIPS tank depletion). Software patches also go through the ops subsystem, as they will change over time and require reconfiguration when bugs are found.

C. Key Operational Requirements

The key operational requirements have been designed to reflect all the mission needs in the final design of the operational systems. Operational needs vary throughout the different mission phases and will focus on logistically and technically providing spacecraft oversight, commandability, and anomaly response capabilities.

Ground Segment Requirements include the following.

1. The ground segments shall maintain the capability to command and receive telemetry from the spacecraft whenever geometric line-of-sight to Earth exists.

2. The ground segments shall support an average of at least one effective contact window at any given moment, enabled through global ground stations.

3. The ground segments shall coordinate ground station scheduling such that no single ground asset is a single point of failure for mission-critical commanding.

4. The ground segments shall support 24/7 link availability during nominal science operations, subject to atmospheric conditions at optical ground stations.

5. The operations system shall maintain 24/7 staffing coverage during Launch and Early Operations Phase (LEOP), the Jupiter gravity assist, the solar perihelion maneuver, and the first science acquisition at 650 AU.

6. The ground segments shall maintain 8/5 staffing coverage during nominal cruise phases, with on-call anomaly response capability. The ground segments shall maintain anomaly response capability within 12 hours during cruise and within 2 hours during critical events.

7. The ground segments shall maintain archived copies of all science and housekeeping data at the Mission Operations Center.

8. Atmospheric sensors shall continually support ground stations during nominal operations to enable data-informed planning and station monitoring.

Flight Operations Requirements include the following.

1. The flight operations system shall support a maximum command turnaround time of < 11 days, accounting for two-way light-time delay at distances approaching 950 AU.

2. The flight operations system shall support delayed command acknowledgment due to light-time and shall not rely on real-time operator intervention for nominal maneuver execution.

3. The flight operations system shall support autonomous execution of trajectory correction maneuvers, attitude realignment for optical downlink, and transitions between cruise, science, and high-rate downlink modes.

4. The flight operations system shall allow for uplink correction of spacecraft clocks and ephemeris solutions as required.

Due to the extreme distances, traditional LEO considerations such as passes and contact duration during passes are irrelevant. Data latency and command turnaround time are also not particularly relevant to primary operational concerns. Because the spacecraft is so autonomous, operations will be primarily focused on data acquisition and anomaly detection.

II. Operations Architecture

A. Ground Segment Elements

The ground segment for the SGL mission is designed to support continuous communication with a spacecraft operating at unprecedented heliocentric distances of 650–900 AU. Given the mission's reliance on infrared optical communications and the multi-day light-time delays, the ground architecture emphasizes redundancy, global coverage, and robust data handling capabilities.

1. Mission Operations Center (MOC)

The Mission Operations Center serves as the central hub for all spacecraft commanding, telemetry processing, and mission planning activities. Located at a primary facility (nominally at JPL or an equivalent deep-space operations center), the MOC houses the flight operations team and provides the following capabilities:

- **Command Generation and Validation:** All telecommands are developed, simulated, and validated at the MOC before uplink. Given the 3.75–5.20 day one-way light time during science operations, commands are time-tagged and uploaded as activity blocks rather than real-time instructions.
- **Telemetry Monitoring and Analysis:** Real-time (delayed) telemetry display, limit checking, and trend analysis for all spacecraft subsystems. Automated alerting systems notify operators when parameters exceed predefined thresholds.
- **Mission Planning and Scheduling:** Development of short-term (weekly) and long-term (monthly/yearly) activity plans, coordinating spacecraft modes, maneuvers, science observations, and downlink sessions.
- **Anomaly Response Coordination:** The MOC serves as the decision-making center for anomaly investigation and recovery procedure development.

2. Ground Stations and Data Networks

The SGL mission employs four globally distributed ground stations optimized for infrared optical communications. Unlike traditional RF-based deep space networks, optical ground terminals require clear atmospheric conditions, driving the selection of high-altitude, arid locations:

Ground Station	Longitude (°E)	Primary Advantage
Fucino Space Center, Italy	13.6	European coverage; existing infrastructure
West Australia Space Center	115.4	Arid climate; minimal atmospheric interference
Haleakala Observatory, Hawaii	203.8	Above cloud layer; excellent seeing conditions
Atacama Observatory, Chile	292.3	High altitude; minimal precipitation

Table 4 SGL mission optical ground station network.

The four-station architecture provides the following:

- **Continuous Coverage:** With longitudinal separations of approximately 90°, at least one station maintains line-of-sight to the spacecraft at any given time.
- **Weather Redundancy:** The fourth station (beyond the minimum three required for 24/7 coverage) ensures that localized atmospheric conditions at any single site do not interrupt mission-critical communications.
- **Data Routing:** All stations connect to the MOC via secure terrestrial fiber networks, with automatic handoff protocols as the spacecraft moves between station visibility windows.

Each ground station is equipped with optical terminals capable of both uplink (command transmission) and downlink (telemetry/science data reception) at the mission's specified data rates. Atmospheric sensors at each site continuously monitor seeing conditions to support predictive scheduling.

3. Flight Dynamics and Mission Planning Tools

The Flight Dynamics facility, co-located with or networked to the MOC, provides trajectory analysis, maneuver planning, and navigation support throughout the mission. Key tools and functions include:

- **Trajectory Design and Optimization:** High-fidelity propagation software models the Earth–Jupiter–Sun–650 AU trajectory, accounting for planetary ephemerides, solar radiation pressure (particularly during solar sail deployment), and relativistic effects at solar perihelion.
- **Maneuver Planning:** Computation of Δv requirements for the Trans-Jupiter Injection, Jupiter Oberth maneuver (1.12 km/s at 1.0 R_J), and Solar Oberth maneuver (36.34 km/s at 0.046 AU perihelion). Maneuver sequences are validated against propellant budgets and spacecraft constraints before command generation.
- **Navigation and Orbit Determination:** Processing of radiometric tracking data (range, Doppler) and optical navigation observations to refine spacecraft state estimates. During the science phase, navigation accuracy supports the axial maneuvers required to compensate for stellar wobble of Proxima Centauri b.
- **Ephemeris Updates:** Periodic uploads of updated ephemeris data to the spacecraft OBC to maintain accurate onboard time and position knowledge, critical for autonomous pointing and time-tagged command execution.

4. Data Processing and Archiving Systems

Given the multi-decade mission duration and the scientific value of SGL imagery, robust data processing and archiving infrastructure is essential:

- **Level 0 Processing:** Raw telemetry frames received from ground stations undergo frame synchronization, error correction, and time-tagging before storage.
- **Level 1–2 Processing:** Housekeeping telemetry is converted to engineering units and checked against operational limits. Science data packets are unpacked and organized by observation sequence.
- **Level 3 Processing:** Reconstructed and calibrated science products, including deconvolved imagery of Proxima Centauri b from Einstein ring observations, are generated for distribution to the science team.
- **Long-Term Archive:** All raw and processed data are stored in redundant, geographically distributed archives with a retention period extending well beyond the nominal 39.43-year mission duration. Metadata catalogs support efficient retrieval for future analysis.

The data processing pipeline is designed to operate with minimal human intervention during nominal operations, consistent with the reduced staffing model during cruise and science phases.

B. Operational Data and Command Flow

The SGL mission adapts a closed-loop telemetry architecture designed to support operations at heliocentric distances exceeding 650 AU. The telemetry will originate onboard the spacecraft from all major subsystems, including ADCS, OBDH, COM, EPS, thermal, and payload. The data will be essentially packed and formatted using CCSDS protocols and will be prioritized based on mission criticality, and this will be stored in mass memory during non-contact periods. We know that during the scheduled communications windows, telemetry is transmitted through a high-gain antenna to the designated ground segment, which is usually through the NASA Deep Space Network. The ground station itself performs RF reception, demodulation, frame synchronization, and error correction before sending the data through terrestrial networks to the Mission Operations Center (MOC). At the MOC, telemetry will undergo processing from raw frame (level 0) to unit conversion and limit checking (level 2) to reconstruction and calibration (level 3). The processed telemetry is distributed to the subsystem and will be archived in the long-term storage systems. This is done to support trending, anomaly investigation, and scientific analysis. Due to light-delays, the spacecraft telemetry must be comprehensive enough to support autonomous assessments between ground contacts.

The Telecommand generation for our SGL mission starts at the Mission Operations Center, where the spacecraft activity plans, maneuver sequences, campaign updates, and anomaly recovery procedures are developed. The commands themselves will undergo deep validation, simulation testing, safety validation, and finally approval before any uplink. Since we may expect extreme communication delays common in SGL operations, most commands will be time-stamped and uploaded as some form of activity blocks rather than normal interactive instructions. Once these are validated, the Telecommand frames are transmitted through a secure terrestrial link to the designated ground station and then up-linked trough the deep space network to the spacecraft itself using the communication windows. Once it is received, the spacecraft will perform error detections, authentication of commands if implemented, and some buffering before execution. The commands may be executed immediately or at set onboard times, but the execution status is reported in subsequent telemetry frames. In order for us to be able to receive confirmation of command success, the spacecraft must autonomously be able to detect invalid or unsafe command conditions and also implement fallback or "safe mode" procedures without the ground station having to intervene. This is to ensure continuous mission safety and smooth operations.

C. Interfaces with Spacecraft Subsystems

The Operations subsystem is very important to our mission since it connects people, hardware, and software on the ground to our spacecraft's onboard subsystems in deep-space. This subsystem is responsible for command generation, telemetry monitoring, and mode transitions being executed in a controlled manner. Each of our subsystem interfaces needs to be explicitly structured so that it can support a command execution that is timestamped, along with autonomous fault detection and delayed telemetry verification in our deep-space mission with significant light-time delays. This generalized approach is very similar to other deep-space missions from the past, like the New Horizons mission. In the New Horizons mission, their subsystem integration was largely driven by how the spacecraft had to support itself and its systems for a long time in the deep-space environment between its command and data handling, telecommunications, attitude control, and power management [6].

1. Interface with OBDH

The operations subsystem interfaces with the Onboard Data Handling (OBDH) subsystem through command generation, time synchronization, and telemetry analysis. All of the telecommands that are generated by the Mission Operations Center (MOC) are validated on the ground and are timestamped relative to the time on the spacecraft. Once this is achieved, they are then uploaded in sequence blocks to undergo autonomous execution. The OBDH will then store each of these sequences, execute them according to the onboard clocks, and then report the execution status through telemetry packets. The housekeeping telemetry includes things like processor health, memory usage, watchdog status, RTG output, and subsystem state flags. The housekeeping telemetry will be continuously downlinked during nominal operations. Operations personnel would perform a trend analysis to monitor any form of degradation over the multi-decade mission lifetime. In the event of anomaly detection by onboard Fault Detection, Isolation, and Recovery (FDIR) logic, our OBDH will autonomously reconfigure the spacecraft and report the event during the next communication window. The Operations will retain supervisory authority for most long-term configuration changes. However, it will not rely on real-time commanding due to the distances involved and the multi-day round-trip light time.

2. Interface with Communications (COM)

The interface between the Operations and the Communications subsystem is governed mainly by scheduled downlink windows, power constraints, and pointing geometries. During the nominal cruise phases, a low-rate housekeeping link will be maintained continuously for tracking and health monitoring. High-rate science downlink sessions will be scheduled during dedicated windows and will require coordinated transitions to Downlink Mode. The Operations subsystem needs to ensure that our spacecraft time is synchronized with ground time prior to executing any sequences that are deemed to be time-critical. The power margin needs to be verified before initiating high-rate transmission. The attitude stability must also meet optical terminal pointing requirements. The optical communications require a microradian-level pointing accuracy. For this reason, the Operations subsystem must coordinate closely with ADCS so that it can verify attitude convergence before enabling any sort of high-rate transmission. If our link margin falls somehow below the threshold or thermal limits are approached during the sustained transmission, the Operations must perform command throttling or terminate the downlink session.

3. Interface with ADCS

The interface between the Operations subsystem and the ADCS subsystem is very critical for both science imaging and optical communications. The Solar Gravitational Lens mission requires high-precision pointing for alignment with the Einstein ring, as well as stable Earth-pointing during our downlink. The Operations commands mode will transition between Cruise, Science Alignment, and Downlink modes, where each of them will impose its own distinct pointing and stability requirements. Before entering the science mode, Operations will verify the reaction wheel momentum margins, the star tracker health and solution stability, and finally the absence of thermal bias in the precision sensors. The attitude maneuvers will be scheduled as dated sequences and will be executed autonomously. Telemetry confirmation of the achieved pointing error and rate stabilization will be required before our payload acquisition or high-rate transmission begins. In the event that we encounter any sort of excessive jitter or wheel saturation, the ADCS FDIR logic may reduce maneuver aggressiveness or trigger the activation of a Safe Mode. This will, of course, be after Operations performs a root-cause analysis via telemetry review.

4. Interface with EPS

The Operations subsystem will interface with the Electrical Power Subsystem (EPS) to enforce load prioritization and maintain a positive energy balance throughout the mission lifetime. Since RTG output degrades over time, Operations must account for a reduced end-of-life power margin when they are scheduling high-demand activities. Before entering Downlink Mode or entering onboard reconstruction phases that are quite compute-intensive, the Operations subsystem will attempt to verify the available RTG output, the state of charge of the battery, the depth-of-discharge margin, and any thermal constraints that are associated with power dissipation. In any emergency scenarios that we may deal with, like undervoltage or excessive battery discharge, the EPS may autonomously shed non-critical loads and enter a Safe Mode. The procedures of the Operations subsystem are structured in a manner to accommodate such events by prioritizing OBDH, thermal survival, and communications over payload operations.

III. Operation Phases

A. Launch and Early Operations Phase (LEOP)

The Launch and Early Operations Phase (LEOP) encompasses the critical period from spacecraft separation through initial stabilization and checkout, typically spanning 72–96 hours. During LEOP, the spacecraft transitions from a passive, stowed configuration to a fully operational state capable of executing the Trans-Jupiter Injection maneuver. This phase demands 24/7 staffing coverage and continuous ground station contact.

1. Mission Timeline: Launch to LEOP Completion

Time (L+)	Event	Critical Actions
L+0:00	Launch	Launch vehicle ascent and GEO insertion
L+0:45	Separation	Spacecraft separation from upper stage
L+0:50	First Signal Acquisition	DSN/optical terminal acquisition of downlink
L+1:00	Sun Acquisition	Coarse attitude stabilization; solar reference
L+2:00	Initial Health Assessment	Bus telemetry review; EPS, thermal, OBDH status
L+4:00	Antenna Deployment	High-gain antenna deployment and checkout
L+8:00	Full Attitude Convergence	Star tracker acquisition; fine pointing established
L+12:00	Subsystem Checkout Begin	Systematic verification of all bus subsystems
L+48:00	Propulsion System Checkout	Valve cycling; tank pressure verification
L+72:00	Communications Checkout Complete	Uplink/downlink verification at all data rates
L+96:00	LEOP Complete	Spacecraft declared ready for commissioning

Table 5 LEOP timeline and major milestones.

2. First Signal Acquisition and Health Assessment

Immediately following separation from the launch vehicle, the spacecraft autonomously initiates a pre-programmed safe mode sequence:

1) **Detumble and Sun Acquisition:** Using coarse sun sensors and reaction wheels, the spacecraft arrests any residual tip-off rates from separation and establishes a thermally safe, power-positive attitude with solar panels (if applicable) or RTG radiators properly oriented.
2) **Beacon Transmission:** A low-rate RF beacon is transmitted omnidirectionally, allowing ground stations to acquire the spacecraft regardless of attitude uncertainty.
3) **Initial Telemetry Downlink:** Upon ground acquisition, essential housekeeping telemetry is downlinked, including:
   - EPS status: RTG output voltage/current, battery state of charge
   - Thermal status: Critical component temperatures (OBC, propulsion lines, payload)
   - ADCS status: Attitude rates, sun sensor readings, reaction wheel speeds
   - OBDH status: Processor health, memory integrity, watchdog status
4) **Anomaly Screening:** Flight controllers compare received telemetry against predicted values and red/yellow limits. Any out-of-limit conditions trigger immediate investigation per contingency procedures.

3. Deployment Sequence

Following initial health verification, deployable elements are commanded in a controlled sequence to minimize risk:

1) **High-Gain Antenna Deployment:** The primary optical communications terminal and/or RF high-gain antenna are deployed and verified. Deployment is confirmed via telemetry indicators (microswitch status, motor current signatures) and, where possible, by commanding a test transmission and verifying link margin improvement.
2) **Sensor Booms/Appendages:** Any deployable sensor booms (e.g., magnetometer) are released and verified.
3) **Solar Sail Stowage Verification:** The solar sail remains stowed during LEOP; however, stowage integrity is verified via thermal and mechanical telemetry to ensure readiness for deployment at solar perihelion.
4) **Telescope Cover:** If applicable, the telescope protective cover status is verified as closed and secure for the cruise phase.

Each deployment is preceded by a go/no-go assessment based on current spacecraft state and followed by a comprehensive telemetry review before proceeding.

4. Communications and Subsystem Checkout

With deployments complete, systematic checkout of all spacecraft subsystems is conducted:

- **Communications Subsystem:** End-to-end verification of uplink command reception and downlink telemetry at all operational data rates. Optical terminal pointing calibration using known stellar references.
- **ADCS:** Star tracker initialization and alignment verification. Reaction wheel performance characterization. Gyroscope calibration. Verification of pointing accuracy and stability against requirements.
- **EPS:** RTG output trending to establish baseline degradation model. Battery charge/discharge cycling if applicable. Load switching verification.
- **Propulsion:** Thruster valve integrity checks (non-firing). Propellant tank pressure and temperature verification. Line heater functionality.
- **Thermal:** Heater circuit verification. Thermal balance assessment against predictions.
- **OBDH:** Memory load verification. Software version confirmation. Watchdog and safe mode trigger testing.
- **Payload:** Initial power-on and health check of science instruments (telescope, detectors). No science operations conducted; focus is on verifying nominal telemetry.

5. Initial Orbit Determination

Although the spacecraft is inserted directly into a heliocentric transfer trajectory rather than a parking orbit, initial orbit determination (OD) is critical for planning the Trans-Jupiter Injection:

- **Tracking Data Collection:** Range and Doppler measurements are collected from multiple ground stations over the first 24–48 hours.
- **State Vector Estimation:** Flight dynamics processes tracking data to determine the spacecraft's position and velocity with sufficient accuracy to design any required trajectory correction maneuvers (TCMs).
- **TCM Planning:** If the achieved trajectory deviates from nominal, a small correction maneuver is computed. The maneuver sequence is validated via simulation before uplink.

6. LEOP Success Criteria

LEOP is declared complete when the following criteria are satisfied:

1) Spacecraft attitude is stable and within pointing requirements.
2) All deployables are verified deployed and functional.
3) Two-way communication is established at operational data rates.
4) All bus subsystems report nominal health with adequate margins.
5) Initial orbit determination confirms trajectory within acceptable bounds for the Jupiter transfer.
6) No unresolved anomalies that would prevent transition to commissioning.

Upon LEOP completion, the spacecraft transitions to the Commissioning Phase for detailed performance characterization and preparation for the Trans-Jupiter Injection maneuver.

B. Commissioning Phase

The commissioning phase begins once the spacecraft has successfully completed LEOP and is stable in power, thermal balance, communications, and attitude control. The goal during this phase is to carefully transition from basic survival capability to full mission readiness. Each spacecraft subsystem is checked in detail to confirm that it is operating within expected limits and that adequate performance margins exist. The power system is monitored for steady RTG output and battery health, and the propulsion system is verified for proper valve and line integrity. Finally, the ADCS system in refined through reaction wheel tuning and star tracker alignment updates to achieve the pointing precision required for future science operations.

After the spacecraft bus is fully validated, attention shifts to the payload. Any deployment mechanisms are commanded gradually with telemetry reviewed after each step to reduce risk. The telescope is optically calibrated using known star fields, and detector performance is characterized to understand noise levels, thermal behavior, and radiation effects. End-to-end tests are performed to confirm that data can be collected, processed onboard, stored, and successfully downlinked. The commissioning phase concludes once performance metrics meet mission requirements and the spacecraft demonstrates reliable autonomous execution, allowing it to enter long-duration cruise operations with confidence.

C. Routine Operations Phase

The routine operations phase itself is broken into the pre-science and mid-science sections, as we will need to travel to 650 AU in $\tilde{3}0$ years before we are able to activate the payload and receive data. As a result, the execution of planned experiments will not commence until the final 10 years of the mission. Once all operations between bus and payload are set, there will be continuous data collection from the telescope, and image processing from the OBC being performed. The downlink for the payload has a cap at 5 KB/s, however it is expected to be substantially lower, allowing for the remaining bandwidth to be picked up by necessary bus telemetry. This extra bandwidth for the bus may go towards special health monitoring, through the use of dwell TLM, which is reserved faster refresh for TLM that we wish to observe at a faster rate than typical TLM. The standard nominal TLM speed will be a major frame every 128 seconds, which allows multiple passes of more critical TLM in a single major frame and is useful for catching issues as soon as possible. The importance of fast feedback from the spacecraft is due to the long transfer time between the ground and the s/c, which is over 5 days by the start of the science phase. When issues are found, updates and maintenance will be performed, then sent to the s/c through up-link commanding. The management of these issues, anomalies, and bugs are found through irregular and unexpected TLM, ranges, and FDIR trips. Careful planning from the Flight Dynamics team will need to be done to have sensitive understandings of expected nominal conditions, and fast response to issues will need to be performed. This will happen through the use of autonomous OBC response, which has been shown to be possible through the use of machine learning [5].

IV. Operational Procedures and Timelines

A. Procedures

The procedures of the Operations subsystem are well structured and define step-by-step actions needed to be taken to achieve our mission objectives while also maintaining the safety of the spacecraft. For a deep space Solar Gravitational Lens (SGL) mission with a multi-day round-trip light time, our procedures must be highly deterministic. This means that they must be aware of the power consumed and the compatibility of time-tagged command executions. Since real-time commanding is infeasible at great distances approaching 650-900 AU, the majority of spacecraft activities will be executed autonomously through pre-validated command sequences. These procedures stay very consistent with previous deep-space operations in the sense that dated sequences are developed, simulated, validated, and uplinked in advance of execution [7].

1. Nominal Procedures

Nominal procedures govern routine mission phases that include cruise operations, science data acquisition, and scheduled downlink sessions. During our extended cruise, the spacecraft will operate in a low-activity mode that emphasizes thermal stability, preservation of power margin, and continuous housekeeping telemetry. The timestamped maneuver sequences will be uploaded in advance and executed autonomously to try to maintain trajectory and pointing geometry. Before executing any spacecraft maneuvers, the onboard logic will verify our reaction wheel momentum margins, battery state of charge, and RTG power availability. The telemetry packets that confirm our maneuver completion and subsystem health will be downlinked during the nominal communications window. We will also include memory scrubbing and periodic health checks to mitigate any sort of errors caused by radiation. Once we have reached the SGL focal region, the operations will transition the spacecraft to Science Mode through a prevalidated command sequence. Some preconditions include a confirmed attitude convergence within pointing tolerance, stable star tracker solutions, and adequate thermal and power margins. The payload will then perform photon integration and onboard reconstruction according to scheduled tasks that are timestamped. Intermediate data products will be stored and verified before the downlink. If by any chance our pointing error exceeds allowable limits during the acquisition phase, the procedure will branch to an attitude stabilization subroutine before we resume our science operations. Our downlink sessions will be scheduled based on ground-station geometry and on power availability. Before we attempt to initiate a high-rate transmission, our spacecraft will verify a sufficient RTG output and battery margin so that it can support peak loads. The optical terminal will perform precise body pointing, followed by fine gimbal stabilization. Once we are able to confirm a link lock and an acceptable pointing error, we will begin a high-rate transmission. The telemetry monitoring will continue throughout the duration of the downlink session. This includes amplifier temperature, bus voltage, and pointing stability. If any of these parameters exceed thresholds, the procedure will autonomously throttle the data rate or terminate the transmission so that it can protect spacecraft resources.

2. Contingency Procedures

Contingency procedures are predefined recovery actions for anomalous conditions. Given the long communication latency, the spacecraft must be capable of autonomous anomaly response before any ground intervention occurs. If our bus voltage drops below a pre-defined limit or our battery depth of discharge exceeds an accepted value, our onboard FDIR logic will immediately initiate load shedding. Subsystems that are not that important, like high-rate payload processing or downlink operations, must be disabled first. The spacecraft will transition to Safe Mode if we deem that to be necessary. Otherwise, we would prioritize OBDH, thermal survival, and our low-rate communications. If any excessive jitter, wheel saturation, or star tracker faults are detected, the spacecraft must suspend the acquisition of payloads and must execute an attitude stabilization routine. If our stabilization somehow fails within a set period of time, the Safe Mode will be activated, and our fault summary will be prepared for any transmissions. Also, if the temperatures of any critical components onboard our spacecraft approach or exceed operational limits during activities that require high amounts of power, our procedure will automatically command a reduction in computational load or transmission power. For conditions involving overtemperature during our downlink phase, the spacecraft will terminate any incoming transmissions and will return to a Nominal Mode so that it can reestablish thermal equilibrium.

B. Timeline Generation

Because of the extreme communication delays associated with operating hundreds of astronomical units from Earth, all timelines are developed on the ground and uploaded as time-tagged command sequences. Activities are planned well in advance and organized so that spacecraft safety always takes priority. Core functions such as thermal control, power regulation, and attitude stability are protected first, while science operations and high-rate downlink sessions are scheduled only when sufficient resource margins exist. Before a timeline is approved, it is reviewed to ensure that power demand, pointing requirements, thermal limits, and onboard storage capacity remain within safe bounds.

Ground station availability strongly shapes the operational schedule. Communication passes are reserved in advance, and command updates are bundled for uplink during those windows. Commands are created in UTC and converted to spacecraft event time, accounting for light-time delay and onboard clock drift. During the science phase, operations generally follow a repeating weekly cycle. For example, Days 1 through 4 may consist of continuous autonomous payload observations and onboard image processing while housekeeping telemetry is stored. On Day 5, the spacecraft transitions to a communications attitude and performs a scheduled commanding window, receiving updated sequences and clock corrections. Day 6 may include a dedicated high-rate downlink period in which stored science data are transmitted to Earth, followed by health checks and momentum management. On Day 7, the spacecraft returns fully to science attitude and resumes data collection. This structured and repeatable planning approach allows the mission to operate efficiently and safely despite the inability to intervene in real time.

V. Staffing, Automation, and Anomaly Response

A. Staffing Concept

The SGL spacecraft will be heavily autonomous throughout the duration of the mission. This means the staffing will be mitigated during a significant portion of the mission, however, an operations team will still be required for daily surveillance and will require more attention during the more important maneuvers and phases. Due to the sheer distance the spacecraft will be during the science phase, real-time operating will be impossible. The staffing Will involve at most five people per shift. These will consists of a main flight director, systems engineer, spacecraft operation manager, as well as event-based operators such as ADCS, Propulsion, and Communications. In terms of these roles, the flight director will have overall authority over the mission, the systems engineer will have an indepth understanding of the technical integration of the vehicle and will be able to monitor it. Lastly, the spacecraft operator will execute the commands to the spacecraft and manage the telemetry.

The metrics which decided these phases are measured based off of the risk during various levels of the mission. The mission staffing will consist of three phases. Launch and early operations, as well as the Jupiter and Oberth manuevers, will require 24 hour surveillance, with three shifts a day consisting of all personal, totaling to 18 separate personnel during these stages. The second phase will be the post-solar escape cruise to 650 AU and beyond. Due to the high autonomy of this phase, the 24-hour requirement will shift to a four hour contact window, requiring only the flight director, systems engineer, and operation manager 4-5 days a week. During this phase, the other specified operators will be on call. Finally, during the science phase. This will be similar to the cruise phase, however, a payload scientist will be involved for information with data processing and the like.

B. Automation and Autonomy

Onboard systems will be highly autonomous, only relying on ground stations for clock adjustments and ephemeris uploads. Multi-day communication delays will make manual commanding incredibly difficult after reaching 550+ AU. Time-tagged commands will be the primary way of initiating events and maneuvers at predetermined times during the cruise[8]. Some examples include the trajectory maneuvers for the Earth-Jupiter transfer, solar infall, and hyperbolic escape. Time-tagged commands will ensure mode transfers align with communication and data acquisition opportunities. Optimal times for science gathering and high-rate downlink will be turned into these timed commands. This highly autonomous system will alleviate pressure from the ground stations to constantly monitor and react to onboard telemetry changes. This allows the staffing requirements to drop to a 8/5 system during nominal conditions. A fleshed out monitoring and alerting system with FDIR triggers will protect the spacecraft and react to dynamic conditions like battery levels, thermal extremes, or ADCS pointing [9]. Finally, a Safe Mode/Contingency Mode will put the spacecraft into a state of self preservation in the event of a rare mission-threatening event so that ground teams can react to the event and drive key decision making. This retains ultimate mission authority within ground teams and helps mitigate the dangers of a fully automated mission.

C. Anomaly Management

In order to make sure the mission runs smoothly, any deviations in expected results must be detected and handled quickly before it damages the spacecraft. With most missions, this can be done through uplink with ground control; however, since the crux of this mission takes place light-years away, automatic detection is a necessity. Thankfully, it has been demonstrated by JPL that, by using machine learning, this is possible.[5] The way it works is that the model will attempt to predict certain values, such as temperature or fuel consumption, and if there is a strong deviation from the prediction, it is flagged as an anomaly. This will also be sent to the ground station for verification, though that will take several days. Additionally, there will also be a limit system in place where, if the values exceed a maximum or minimum, the data is flagged as an anomaly.

When an automatic detection occurs, the isolation process will begin by utilizing built-in testing (BIT). There will be 3 individual pieces of radiation-tolerant hardware that will each send a test signal to any suspected components. These signals will have to cover all operations the subsystems may perform in order to catch the error. The subsystem will then send back a response signal to the BIT. After taking a look at the received signal and comparing it to the expected, a majority vote will occur to determine if the component is working or faulty. The reason for the majority vote is that the BIT system can also be faulty. As a result, in order to make sure the validation system is more reliable than the spacecraft, extreme redundancy is utilized in the form of the 3 processors. The result of the process will also be sent to the ground station for them to double-check. This should ideally be done completely automatically without human intervention.

After the fault has been isolated to a certain component, one of two recovery mechanisms may happen, depending on the subsystem:

1. The cold redundant system is turned on

This applies in situations where there is a cold redundancy, like for the payload processing, main bus, or BIT system. The defunct copy will automatically turn off, and operation will continue with the new one. This data will get sent to the ground station for confirmation. If both copies have issues, then an attempt will be made to revitalize the processors using the second method.

2. An alternate circuit pathway is used

For systems that do not have a backup, or in situations where all redundant systems fail, an attempt to go around a faulty circuit using a different pathway is made. The hope is that the data can still be transferred at the cost of a lower resolution or rate. Additionally, there will be backup pathways for some time-critical operations (such as the ADCS system) that will automatically activate following detection and isolation. For critical operations that are not time sensitive, the code that causes this change will be updated using uplink from the ground station. Additionally, if one subsystem begins to fail, processors from another subsystem could attempt to take up some of the work. For example, if the propulsion control circuit is faulty, then code can be sent up that uses some of the processing power of the FDIR system to control propulsion instead.

VI. Operations Risks and Verification

A. Key Operational Risks

Although the same anomalies are able to be solved using the FDIR system, there are critical failures that have to be avoided at all costs, as they run a massive risk of completely killing the mission. For this spacecraft, a list of the key operational risks is provided below. Note that for many of these, critical means losing the ability to communicate with the ground station, or the problem can not be solved via any means.

1. Failure of the ADCS subsystem

The ADCS subsystem has no cold redundancy. This means that if it were to fail, and the backup route were to fail, the mission would be dead since the satellite is likely no longer able to receive updated code from the ground station. Additionally, if the system were to fail before the solar dive, it would not be able to point the heat shield the correct way, and the satellite would burn up.

2. Failure of the solar sail deployment

Since the solar sail is only really effective during the Sun Oberth, if it fails to be put out in time, the mission will lose a lot of its propulsive ability, and it will take too long to reach the desired destination, deeming the mission a failure.

3. Failure of the telescope deployment

If the telescope fails to deploy, the mission cannot achieve its main goal. Additionally, there would likely be no software update available to fix the issue, as it will be hardware-based. The result is that if the mechanism is jammed, the telescope will never deploy.

4. Failure of the decoder

Most unique failures that are not fixed automatically can be fixed by updating the code from the ground station. However, if the decoder that turns those uplink signals into data readable by the computer fails, it is extremely likely that the mission will fail due to an inability to fix the inevitable errors that will occur.

5. Failure of the encoder

If the encoder were to fail, it would mean that any data produced from the mission would be lost during the defective phase, rendering the satellite useless. However, because the ADCS and decoder are likely still working, code can be sent up to fix the encoder. The problem is that the ground station would have no way of knowing what the problem was and the effects any updated code sent had on the encoder operation. For this reason, the failure of the encoder is deemed a critical risk.

B. Mitigation and Verification

Primarily during preparation phase, but also throughout the rest of the mission, Ops Engineers should perform rehearsals, Day-in-the-Life's, and simulations replicating nominal and contingency scenarios in order to ready themselves for the potential challenges of the mission, as well as to weed out any remaining issues that may have slipped past production. These exercises are useful for mitigating the risk of the mission, and allows for the engineers to be properly trained for the untested in the field engineering that will be performed. Because the mission will take decades, continuous training will need to be performed, since it would be very easy to forget plans and for the changing of the guard to lose their understanding of the software and hardware.

End-to-end (E2E) tests verify that the spacecraft and ground systems work all the way through the chain of operations from start to finish. This means following the line from planning, command generation, uplink, execution, telemetry generation, downlink, ground reception, display, archiving, and analysis [10]. The tests performed to verify E2E include checking timing, encryption, and configuration at each stage of the chain. For each of the procedures created -whether that be for testing, operations, or just standardization- review and validation will need to be performed from a minimum of 3 total engineers qualified to write and validate. This allows for formal approval, accountability, and prevents single point error from operators. Checklists and verification will be implemented into all procedure creation and updating.

VII. Conclusion

The Operations Subsystem architecture for the SGL mission is designed to maintain a reliable, long term deep-space operation at distances beyond 650 AU, a distance that makes real-time control impossible due to the light-time delays. The design selected integrates a main mission operations center, Optical/RF ground stations positioned globally to have nearly continuous coverage, with a closed-loop telemetry and command architecture which will be time-tagged. Although the spacecraft will be highly autonomous, this provides strong ground oversight for key stages of the flight.

Important phases of the flight will involve launch and early operations, critical Jupiter and Solar Oberth maneuvers, extended cruise and science phase. These phases will be supported and validated using a predetermined timeline, fitting structured mode transitions such as cruise, science, downlink, and safe mode. The staffing will scale with the risk in each mission phase, requiring 24/7 coverage during early stages and high-risk events, while being reduced to as little as 4/5 monitoring during cruise and science phase. This is possible with the use of high onboard autonomy, FDIR logic, machine-based learning for anomaly detection, and safe-mode protection. Many of the assumptions made will include reliable RTG power availability, successful solar sail and telescope deployment, and precise ADCS performance for optical communications and science stage. Operational risks include ADCS failure, encoder/decoder faults, and sail or telescope deployment anomalies. These can be mitigated with redundancy, thorough validation, autonomous fault detection and isolation, and repeated system testing. Overall, the design shows a strong framework with the ability to sustain scientific returns across a multi-decade interstellar mission.

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